Transportation in Wales

Introduction:

There are many ways to travel around Wales, including bus, train, airplane at Cardiff airport, taxi, car or cycling. Transportation is used to get around Wales from one area to another. Transportation in Wales has a few issues it needs to overcome that I discovered from customer reviews on transportation. These consist of transportation systems not showing up on their scheduled times, not being enough spaces on them for the customers to sit down or comfortably stand with enough room, or being late and overpriced. This causes challenges for transportation systems because less people will be inclined to use transportation if these problems keep occurring but they need to increase the prices to keep up with the running costs so they go round in a loop of people not using transportation due to issues, rising prices can't go down because not enough people are using it. The data gathered is important and relevant to the research question because it helps us to understand the satisfaction with transportation based on a range of different scenarios. In this report I talk about the data, how it effects transportation use and the challenges and issues transportation face in more detail. I also come up with a plan to over come these issues.

Methodology:

Data pre-processing is where steps are taken in the data mining process, this is where it takes the raw data and transforms it into readable data so it is understood by computers. Data pre-processing is important because as the phrase says "garbage in, garbage out", (Mesevage, 2021) if good, organised and correct data doesn't go in, then good reliable and correct data won't come out and can give bad results.

The first step of the data pre-processing steps involve data capturing and extraction. This is the process of extracting the data and information that is needed to then be used for further analysis. This data can act as a fuel to then help the automated workflow and decision making. Once the data has been extracted it is then cleaned, this means fixing and removing any data that is incorrect, a duplicate set of data or incomplete data, leaving us with accurate and correct data to further use. In the data cleaning cycle there are steps to follow to ensure all the data is correct and there are no mistakes. The first step is to identify the data that is incorrect, this can include typos, duplicate data, missing data or data that is incorrect, then fix the mistake by either deleting the data or fixing it, then set data cleansing techniques and filter the outliers and fix missing data, then the implementation process. This should leave the data accurate and correct. Data integration is the next step and this involves bringing data in from many sources so that it provides a complete and accurate dataset to work with. Benefits of data integration include it being easy and available with fast connections between the different data stores, it provides seamless knowledge transfers between the different systems, it has better collaboration, it provides better data integrity and data quality and it has increased efficiency and ROI. Data integration is important because companies gather large amounts of data daily from all different sources and for this data to be meaningful it must be accurate, and easily accessible for analysis. Next data design is the translation of raw data into an understandable format that allows easy viewing of patterns and trends. The next step is data analysis which involves the deciphering and processing of data, to enable a new meaning that meets the requirements of a businesses logic. After this data visualisation is done and involves the raw data being transformed to a pictorial or graphical form. This allows the data to be easier to read, sometimes due to the volume of data.

To answer the research question I first needed to understand why I needed data analysis and then began collecting the data I thought was relevant to transportation in Wales from the data gathered from the surveys, I cleaned through the unnecessary data and picked out the important data, I then reasoned why I thought this data was relevant and analysed the data and then interpreted the results and applied it to the research question.

Results:

Table 2 - overall satisfaction with state of transport system in Wales, by having use of a car.

Use of a car	Mean
Yes	5.8
No	6.4
Total	5.9

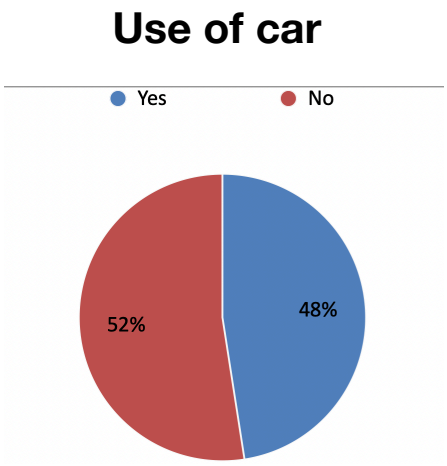


Table 4 - overall satisfaction with state of transport system in Wales, by urban/rural area.

Urban/Rural area	Mean
Urban	6.0
Rural	5.5
Total	5.9

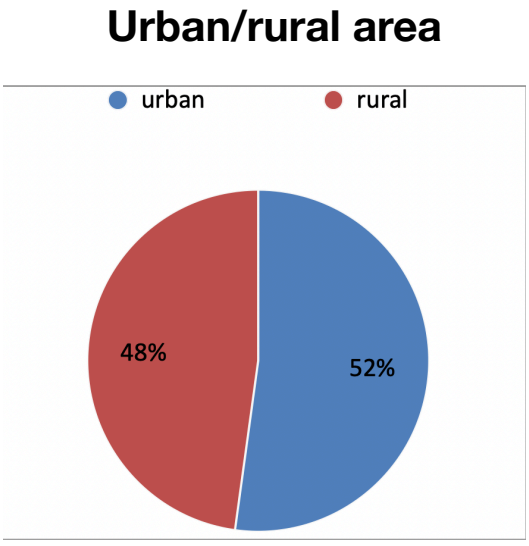


Table 7 - overall satisfaction with state of transport systems in Wales, by WIMD deprivation score.

WIMD deprivation score	Mean
20% most deprived	6.1
20%-40% most deprived	6.0
40%-60% most deprived	5.8
20%-40% least deprived	5.7
20% least deprived	5.8
Total	5.9

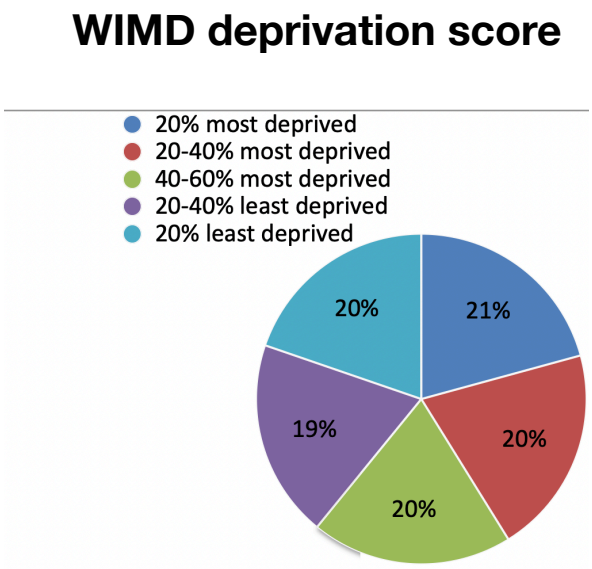


Table 10 - overall satisfaction with state of transport system in Wales, by age.

Age	Mean
16-24	6.2
25-44	5.9
45-64	5.5
65-74	5.9
75 and over	6.5
Total	5.9

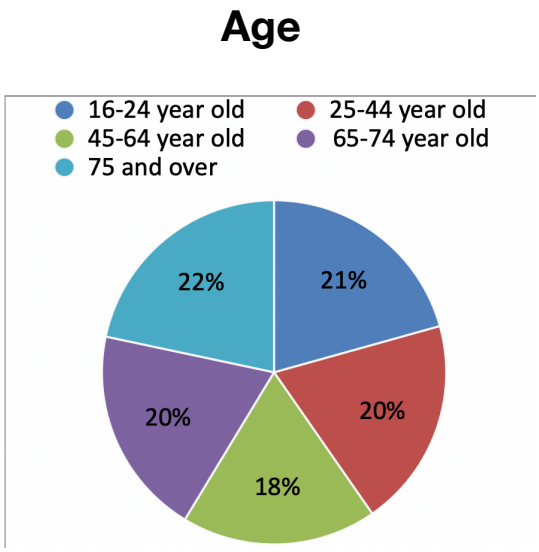


Table 11 - overall satisfaction with state of transport system in Wales, by employment status.

Employment status	Mean
In employment	5.7
Not in employement	6.1
Total	5.9

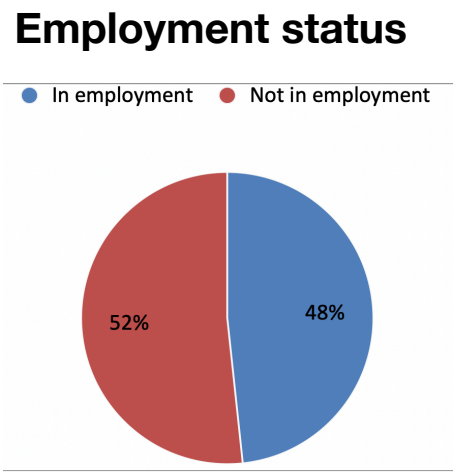


Table 13 - overall satisfaction with state of transport system in Wales, by local area is free from heavy traffic.

Local area is free from heavy traffic	Mean
Agree	6.0
Disagree	5.8
Neither agree nor disagree	5.7
Total	5.9

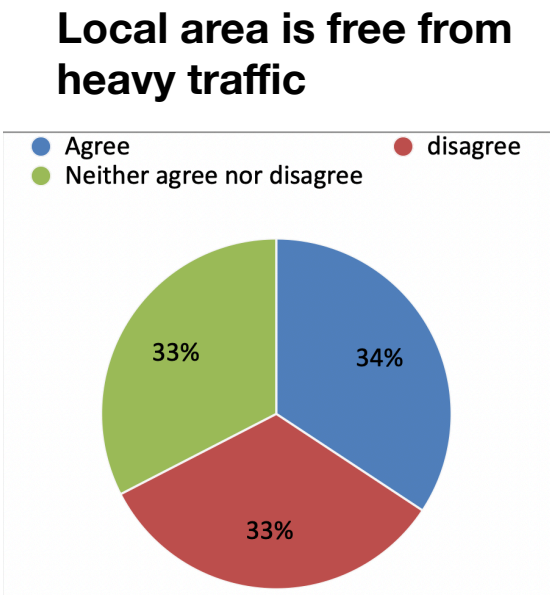


Table 14 - overall satisfaction with state of transport system in Wales, by feeling of safety travelling after dark.

Feeling of safety travelling by public transport after dark	Mean
Very safe	6.2
Fairly safe	5.9
Fairly unsafe	5.5
Very unsafe	5.7
Total	5.9

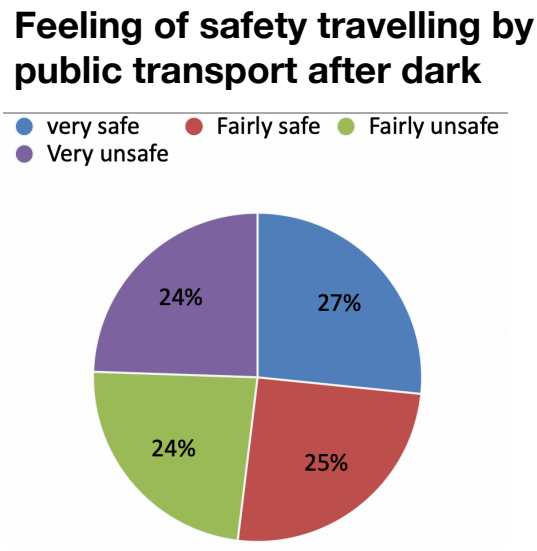
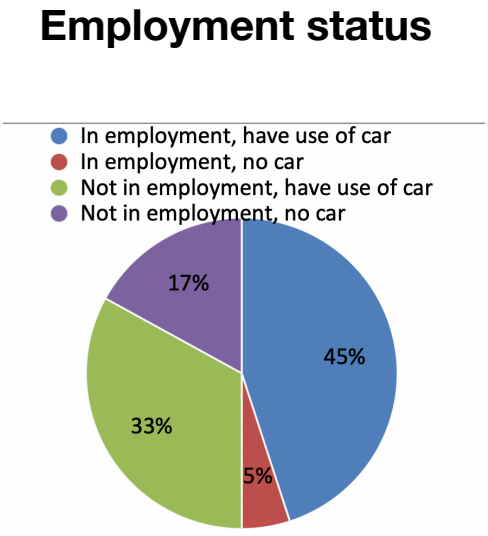


Table 21 - have use of car, by employment status

Employment status	Have use of car (mean)	
	Yes	No
In employment	90%	10%
Not in employment	66%	34%
Total	79%	21%



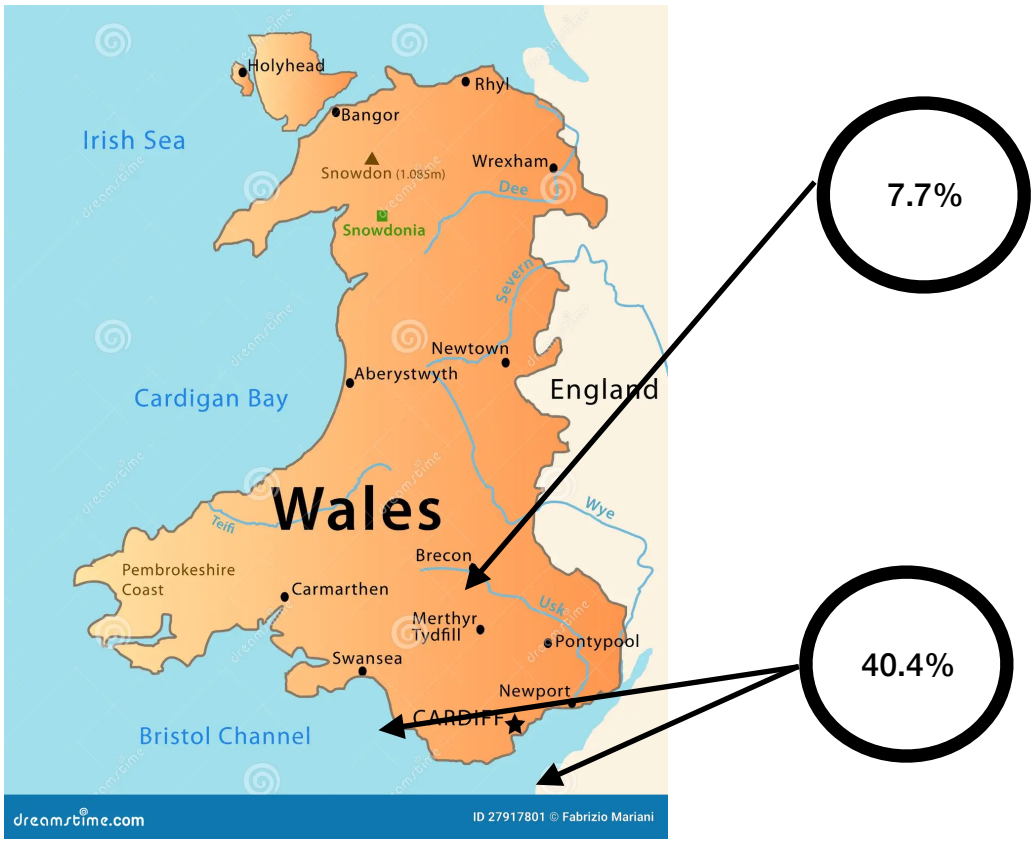
Data taken from national survey for Wales results, 2013-14: Transport.

Discussion:

The surveys above were conducted on a range of people to help gather an idea of the current issues of transportation in Wales. I chose to show the data as tables and as pie charts because I thought this was the best layout for me to clearly see and compare the data and the pie chart allowed me to see the percentages of each table to compare them.

The tables I picked to analyse, I picked because I feel like they show relevance to the issues relating directly to transportation. The use of a car is important as cars are a form of transportation so knowing how many people use them is important to factor in to the issues relating to transportation. Less people have a car at 48% than those who don't have a car at 52% meaning more of the population with be using other means of transport. The second table I picked was the overall satisfaction of transportation by rural or urban area and this is important to get an understanding of the areas that might have the most issues with transportation. With urban area having a higher satisfaction rate at 52% than rural at 48%.

The highest traffic areas of Wales are Swansea and Cardiff at 40.4% total traffic in Wales. The lowest traffic areas of Wales are Blaenau gwent and merthyr tydfil at 7.7% total traffic in Wales. As shown below on the map.



F, Mariani, (2023). *Wales map*. Available at: https://www.dreamstime.com/stock-image-wales-map-image27917801 [Date accessed 28th October 2023].

Another table I picked was overall satisfaction of transportation based on deprivation score, I picked this because it is important to understand the effects of transportation on those who are most and least deprived as transportation costs money and as some people have complained is overpriced. So 20% most deprived are the most satisfied with transportation at 21% and 20-40% least deprived being the most unsatisfied at 19%. I also picked the table of overall satisfaction based on age, this is because there are discounts for buses for certain age groups and there might be age groups that favour a certain type of transportation so with 75 and over being the most satisfied with transportation I think this relates to the discount for seniors on bus passes, and 45-64 year olds being the most unsatisfied at 18%, because as there is a discount for seniors there is also a discount for youths so this age group has to pay full price and doesn't qualify for discounts on transportation systems. I also picked overall satisfaction with transportation systems based on employment status, I believe this is relevant because transportation can effect employed people by not being on time or not turning up as they have a job to get to on time but it can also effect the unemployed because it means they are more likely to use it more throughout the day and multiple times. The unemployed are more satisfied with transportation at 52% than those employed at 48%. Another tables relates to the area being free of traffic, I think this is relevant because if there is busy traffic it could cause issues with transportation systems. Most people at 34% agree their local area is free from heavy traffic. I picked feeling of safety after dark by transportation because I think this relates to if people don't feel safe they won't use transportation systems. The highest percentage do feel very safe at 27% and the lowest percentage at 24% feel very unsafe. The last table I picked is a mix of employment status and the use of

a car, I think this relates to transportation because employed people with the use of car will use cars at specific times of the day and can cause heavy rush hour traffic during work hours and those without the use of a car will use other transportation systems. The highest percentage at 45% is those in employment with the use of a car so this will contribute highly to heavy traffic. The lowest percentage at 5% is in employment but don't use a car.

I think these tables all relate directly to the issues related to transportation. The issues of transportation include transportation not showing up, being late, not enough seats or space and being overpriced.

After gathering all of the data and reviewing this alongside customers reviews to help establish the issues transportation face I believe the best way to overcome these issues is to first overcome the issues relating to the transportation itself so add more carriages to ensure there space for more people to fit, and then update the transportation timetable so all timetables are up to date and the same and ensuring the transportation systems turn up as they should, keeping the online timetables up to date or delays or transportation that may not be showing up. To overcome these issues first I believe this will encourage more use of transportation systems and this should encourage the price of transportation systems to drop as more people will be using them. There is some difficulty when it comes to overcoming these issues though as adding new carriages does cost money so the transportation systems will be out of pocket to do this, and also to update the timetables this will take someone to dedicate some time and money on working through these timetables and ensuring they are correct which can be difficult.

Conclusion:

To help improve the transportation in Wales after reviewing and discussing the data chosen, why it was important and the relevance to transportation, along with resident and customer views on the issues relating to transportation, I believe to overcome these issues we must first solve the transportation timetable and transportation being late or not showing up as the first step to eradicating the issues, the second step would be to then reduce the cost as more passengers continue to use transportation systems again. To fix these issues it does come with its challenges as described but with hope of a good outcome in the future for transportation in Wales.

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